

Lebsheet-3

1. Write a Java program to input and display 5 integers using a 1-D array.

import java.util.Scanner;

```
public class ArrayInputDisplay {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        // Input 5 integers
        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }

        // Display the integers
        System.out.println("The integers are:");
        for (int i = 0; i < 5; i++) {
            System.out.println(numbers[i]);
        }

        sc.close();
    }
}
```

Output-

Enter 5 integers:

10

20

30

40

50

The integers are:

10

20

30

40

50

2. Create a Java program to input 10 integers and display them in reverse order.

import java.util.Scanner;

```
public class ReverseArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```

int[] numbers = new int[10];

System.out.println("Enter 10 integers:");

// Input 10 integers
for (int i = 0; i < 10; i++) {
    numbers[i] = sc.nextInt();
}

// Display in reverse order
System.out.println("Integers in reverse order:");
for (int i = 9; i >= 0; i--) {
    System.out.println(numbers[i]);
}

sc.close();
}
}

```

Output-
Enter 10 integers:

1
2
3
4
5
6
7
8
9
10

Integers in reverse order:

10
9
8
7
6
5
4
3
2
1

3. Write a Java program to find the sum of all elements in a 1-D array.

```
import java.util.Scanner;
```

```

public class ArraySum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];
        int sum = 0;

        System.out.println("Enter 5 integers:");

        // Input elements
        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
            sum = sum + numbers[i];
        }

        // Display sum
        System.out.println("Sum of all elements = " + sum);

        sc.close();
    }
}

```

Output-

Enter 5 integers:

10

20

30

40

50

Sum of all elements = 150

4. Create a Java program to find the maximum element in an array.

```

import java.util.Scanner;

```

```

public class ArrayMaximum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        // Input elements
        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }
    }
}

```

```

// Find maximum element
int max = numbers[0];

for (int i = 1; i < 5; i++) {
    if (numbers[i] > max) {
        max = numbers[i];
    }
}

System.out.println("Maximum element = " + max);

sc.close();
}
}

```

Output-

Enter 5 integers:

25
10
45
30
15

Maximum element = 45

5. Write a Java program to count even and odd numbers in an array.

```

import java.util.Scanner;

public class EvenOddCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[10];
        int even = 0, odd = 0;

        System.out.println("Enter 10 integers:");

        // Input elements
        for (int i = 0; i < 10; i++) {
            numbers[i] = sc.nextInt();

            if (numbers[i] % 2 == 0) {
                even++;
            } else {
                odd++;
            }
        }
    }
}

```

```

        System.out.println("Number of even numbers = " + even);
        System.out.println("Number of odd numbers = " + odd);

        sc.close();
    }
}

```

Output-

Enter 10 integers:

```

1
2
3
4
5
6
7
8
9
10

```

Number of even numbers = 5

Number of odd numbers = 5

6. Create a Java program to copy all elements from one array to another.

```
import java.util.Scanner;
```

```

public class CopyArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] array1 = new int[5];
        int[] array2 = new int[5];

        System.out.println("Enter 5 integers:");

        // Input elements into first array
        for (int i = 0; i < 5; i++) {
            array1[i] = sc.nextInt();
        }

        // Copy elements to second array
        for (int i = 0; i < 5; i++) {
            array2[i] = array1[i];
        }

        // Display copied elements
        System.out.println("Elements of the second array:");
    }
}

```

```

        for (int i = 0; i < 5; i++) {
            System.out.println(array2[i]);
        }

        sc.close();
    }
}

```

Output-

Enter 5 integers:

10
20
30
40
50

Elements of the second array:

10
20
30
40
50

7. Write a Java program to input and display elements of a 3 × 3 matrix.

```
import java.util.Scanner;
```

```

class Matrix {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] a = new int[3][3];

        System.out.println("Enter elements of 3 x 3 matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.println("Elements of the matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(a[i][j] + " ");
            }
            System.out.println();
        }
    }
}

```

```
}  
}
```

Output-

Enter elements of 3 x 3 matrix:

```
1 2 3  
4 5 6  
7 8 9
```

Elements of the matrix:

```
1 2 3  
4 5 6  
7 8 9
```

8. Create a Java program to find the sum of all elements in a matrix.

```
import java.util.Scanner;
```

```
class MatrixSum {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[][] a = new int[3][3];  
        int sum = 0;  
  
        System.out.println("Enter elements of 3 x 3 matrix:");  
  
        for (int i = 0; i < 3; i++) {  
            for (int j = 0; j < 3; j++) {  
                a[i][j] = sc.nextInt();  
                sum = sum + a[i][j];  
            }  
        }  
  
        System.out.println("Sum of all elements = " + sum);  
    }  
}
```

Output-

Enter elements of 3 x 3 matrix:

```
1 2 3  
4 5 6  
7 8 9
```

Sum of all elements = 45

9. Write a Java program to find the sum of each row in a matrix.

```

import java.util.Scanner;

class RowSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] a = new int[3][3];

        System.out.println("Enter elements of 3 x 3 matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.println("Sum of each row:");

        for (int i = 0; i < 3; i++) {
            int sum = 0;

            for (int j = 0; j < 3; j++) {
                sum = sum + a[i][j];
            }

            System.out.println("Sum of row " + (i + 1) + " = " + sum);
        }
    }
}

```

Output-

Enter elements of 3 x 3 matrix:

1 2 3

4 5 6

7 8 9

Sum of each row:

Sum of row 1 = 6

Sum of row 2 = 15

Sum of row 3 = 24

10. Create a Java program to find the sum of each column in a matrix.

```

import java.util.Scanner;

class ColumnSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

```

```

int[][] a = new int[3][3];

System.out.println("Enter elements of 3 x 3 matrix:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        a[i][j] = sc.nextInt();
    }
}

System.out.println("Sum of each column:");

for (int j = 0; j < 3; j++) {
    int sum = 0;

    for (int i = 0; i < 3; i++) {
        sum = sum + a[i][j];
    }

    System.out.println("Sum of column " + (j + 1) + " = " + sum);
}
}
}

```

Output-

Enter elements of 3 x 3 matrix:

1 2 3

4 5 6

7 8 9

Sum of each column:

Sum of column 1 = 12

Sum of column 2 = 15

Sum of column 3 = 18

11. Write a Java program to transpose a matrix.

```
import java.util.Scanner;
```

```

class MatrixTranspose {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] a = new int[3][3];

        System.out.println("Enter elements of 3 x 3 matrix:");

        for (int i = 0; i < 3; i++) {

```

```

        for (int j = 0; j < 3; j++) {
            a[i][j] = sc.nextInt();
        }
    }

    System.out.println("Transpose of the matrix:");

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            System.out.print(a[j][i] + " ");
        }
        System.out.println();
    }
}

```

Output-

Enter elements of 3 x 3 matrix:

```

1 2 3
4 5 6
7 8 9

```

Transpose of the matrix:

```

1 4 7
2 5 8
3 6 9

```

12. Create a Java program to add two matrices of the same size.

```

import java.util.Scanner;

class MatrixAddition {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] a = new int[3][3];
        int[][] b = new int[3][3];
        int[][] sum = new int[3][3];

        System.out.println("Enter elements of first matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                a[i][j] = sc.nextInt();
            }
        }
    }
}

```

```

        System.out.println("Enter elements of second matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                b[i][j] = sc.nextInt();
            }
        }

        // Adding the two matrices
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                sum[i][j] = a[i][j] + b[i][j];
            }
        }

        System.out.println("Sum of the two matrices:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(sum[i][j] + " ");
            }
            System.out.println();
        }
    }
}

```

Output-

Enter elements of first matrix:

1 2 3

4 5 6

7 8 9

Enter elements of second matrix:

9 8 7

6 5 4

3 2 1

Sum of the two matrices:

10 10 10

10 10 10

10 10 10

13. Write a Java program to multiply two matrices (basic case with 3×3 matrices).

```
import java.util.Scanner;
```

```

class MatrixMultiplication {
    public static void main(String[] args) {

```

```

Scanner sc = new Scanner(System.in);

int[][] a = new int[3][3];
int[][] b = new int[3][3];
int[][] result = new int[3][3];

System.out.println("Enter elements of first matrix:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        a[i][j] = sc.nextInt();
    }
}

System.out.println("Enter elements of second matrix:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        b[i][j] = sc.nextInt();
    }
}

// Multiplication of matrices
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        result[i][j] = 0;

        for (int k = 0; k < 3; k++) {
            result[i][j] += a[i][k] * b[k][j];
        }
    }
}

System.out.println("Product of the two matrices:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        System.out.print(result[i][j] + " ");
    }
    System.out.println();
}
}

```

Output-

Enter elements of first matrix:

1 2 3

4 5 6

7 8 9

Enter elements of second matrix:

9 8 7

6 5 4

3 2 1

Product of the two matrices:

30 24 18

84 69 54

138 114 90

14. Write a Java program to input and display elements of a $2 \times 2 \times 2$ 3-D array.

```
import java.util.Scanner;
```

```
class ThreeDArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][][] a = new int[2][2][2];

        System.out.println("Enter elements of 2 x 2 x 2 3-D array:");

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    a[i][j][k] = sc.nextInt();
                }
            }
        }

        System.out.println("Elements of the 3-D array:");

        for (int i = 0; i < 2; i++) {
            System.out.println("Layer " + (i + 1) + " :");

            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    System.out.print(a[i][j][k] + " ");
                }
                System.out.println();
            }
        }
    }
}
```

Output-

Enter elements of 2 x 2 x 2 3-D array:

1 2 3 4 5 6 7 8

Elements of the 3-D array:

Layer 1:

1 2

3 4

Layer 2:

5 6

7 8

15. Create a Java program to find the sum of all elements in a 3-D array.

```
import java.util.Scanner;
```

```
class ThreeDArraySum {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[][][] a = new int[2][2][2];  
        int sum = 0;  
  
        System.out.println("Enter elements of 2 x 2 x 2 3-D array:");  
  
        for (int i = 0; i < 2; i++) {  
            for (int j = 0; j < 2; j++) {  
                for (int k = 0; k < 2; k++) {  
                    a[i][j][k] = sc.nextInt();  
                    sum = sum + a[i][j][k];  
                }  
            }  
        }  
  
        System.out.println("Sum of all elements = " + sum);  
    }  
}
```

Output-

Enter elements of 2 x 2 x 2 3-D array:

1 2 3 4 5 6 7 8

Sum of all elements = 36

16. Write a Java program to count positive and negative numbers in a 3-D array.

```
import java.util.Scanner;
```

```
class PositiveNegative3D {  
    public static void main(String[] args) {
```

```

Scanner sc = new Scanner(System.in);

int[][][] a = new int[2][2][2];
int positive = 0;
int negative = 0;

System.out.println("Enter elements of 2 x 2 x 2 3-D array:");

for (int i = 0; i < 2; i++) {
    for (int j = 0; j < 2; j++) {
        for (int k = 0; k < 2; k++) {
            a[i][j][k] = sc.nextInt();

            if (a[i][j][k] > 0) {
                positive++;
            } else if (a[i][j][k] < 0) {
                negative++;
            }
        }
    }
}

System.out.println("Number of positive numbers = " + positive);
System.out.println("Number of negative numbers = " + negative);
}
}

```

Output-

Enter elements of 2 x 2 x 2 3-D array:

1 -2 3 -4 5 -6 7 -8

Number of positive numbers = 4

Number of negative numbers = 4

17. Create a Java program to find the maximum element in a 3-D array.

```
import java.util.Scanner;
```

```

class Maximum3D {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][][] a = new int[2][2][2];

        System.out.println("Enter elements of 2 x 2 x 2 3-D array:");

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {

```

```

        for (int k = 0; k < 2; k++) {
            a[i][j][k] = sc.nextInt();
        }
    }
}

int max = a[0][0][0];

for (int i = 0; i < 2; i++) {
    for (int j = 0; j < 2; j++) {
        for (int k = 0; k < 2; k++) {
            if (a[i][j][k] > max) {
                max = a[i][j][k];
            }
        }
    }
}

System.out.println("Maximum element = " + max);
}
}

```

Output-

Enter elements of 2 x 2 x 2 3-D array:

10 25 5 40 15 30 8 20

Maximum element = 40

18. Write a Java program to search for an element in a 3-D array and display its position.

```
import java.util.Scanner;
```

```

class Search3D {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][][] a = new int[2][2][2];

        System.out.println("Enter elements of 2 x 2 x 2 3-D array:");

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    a[i][j][k] = sc.nextInt();
                }
            }
        }
    }
}

```

```

System.out.print("Enter element to search: ");
int search = sc.nextInt();

boolean found = false;

for (int i = 0; i < 2; i++) {
    for (int j = 0; j < 2; j++) {
        for (int k = 0; k < 2; k++) {
            if (a[i][j][k] == search) {
                System.out.println("Element found at position: "
                    + "[" + i + "]" + j + "]" + k + "]");
                found = true;
            }
        }
    }
}

if (!found) {
    System.out.println("Element not found.");
}
}

```

Output-

Enter elements of 2 x 2 x 2 3-D array:

10 20 30 40 50 60 70 80

Enter element to search: 50

Element found at position: [1][0][0]

19. Write a Java program to create and display a jagged array with 3 rows of different lengths.

```

import java.util.Scanner;

class JaggedArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] a = new int[3][];

        // Different lengths for each row
        a[0] = new int[2];
        a[1] = new int[3];
        a[2] = new int[4];

        System.out.println("Enter elements:");
    }
}

```

```

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < a[i].length; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.println("Jagged array:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < a[i].length; j++) {
                System.out.print(a[i][j] + " ");
            }
            System.out.println();
        }
    }
}

```

Output-

Enter elements:

1 2

3 4 5

6 7 8 9

Jagged array:

1 2

3 4 5

6 7 8 9

20. Create a Java program to store marks for students where each student has taken a different number of subjects, then display them.

```
import java.util.Scanner;
```

```

class StudentMarks {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] marks = new int[3][];

        System.out.println("Enter number of subjects for each student:");

        for (int i = 0; i < 3; i++) {
            System.out.print("Student " + (i + 1) + ": ");
            int subjects = sc.nextInt();
            marks[i] = new int[subjects];
        }
    }
}

```

```

// Input marks
for (int i = 0; i < 3; i++) {
    System.out.println("Enter marks for Student " + (i + 1) + ":");

    for (int j = 0; j < marks[i].length; j++) {
        marks[i][j] = sc.nextInt();
    }
}

// Display marks
System.out.println("\nStudent Marks:");

for (int i = 0; i < 3; i++) {
    System.out.print("Student " + (i + 1) + ": ");

    for (int j = 0; j < marks[i].length; j++) {
        System.out.print(marks[i][j] + " ");
    }

    System.out.println();
}
}
}

```

Output-

Enter number of subjects for each student:

Student 1: 3

Student 2: 2

Student 3: 4

Enter marks for Student 1:

80 75 90

Enter marks for Student 2:

85 88

Enter marks for Student 3:

70 82 91 78

Student Marks:

Student 1: 80 75 90

Student 2: 85 88

Student 3: 70 82 91 78

21. Write a program to rotate an array to the left by one position.

```
import java.util.Scanner;
```

```

class LeftRotate {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] a = new int[5];

        System.out.println("Enter 5 elements:");

        for (int i = 0; i < 5; i++) {
            a[i] = sc.nextInt();
        }

        // Store the first element
        int first = a[0];

        // Shift elements to the left
        for (int i = 0; i < 4; i++) {
            a[i] = a[i + 1];
        }

        // Put the first element at the end
        a[4] = first;

        System.out.println("Array after left rotation:");

        for (int i = 0; i < 5; i++) {
            System.out.print(a[i] + " ");
        }
    }
}

```

Output-

Enter 5 elements:

10 20 30 40 50

Array after left rotation:

20 30 40 50 10

22. Count how many times each element appears in a 1-D array.

```
import java.util.Scanner;
```

```

class ElementFrequency {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] a = new int[5];
        boolean[] visited = new boolean[5];
    }
}

```

```

System.out.println("Enter 5 elements:");

for (int i = 0; i < 5; i++) {
    a[i] = sc.nextInt();
}

System.out.println("Frequency of each element:");

for (int i = 0; i < 5; i++) {
    if (visited[i]) {
        continue;
    }

    int count = 1;

    for (int j = i + 1; j < 5; j++) {
        if (a[i] == a[j]) {
            count++;
            visited[j] = true;
        }
    }

    System.out.println(a[i] + " appears " + count + " time(s)");
}
}
}

```

Output-

Enter 5 elements:
10 20 10 30 20

Frequency of each element:

10 appears 2 time(s)
20 appears 2 time(s)
30 appears 1 time(s)

23. Find the second largest element in a 1-D array.

```

import java.util.Scanner;

class SecondLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] a = new int[5];

        System.out.println("Enter 5 elements:");
    }
}

```

```

for (int i = 0; i < 5; i++) {
    a[i] = sc.nextInt();
}

int largest = Integer.MIN_VALUE;
int secondLargest = Integer.MIN_VALUE;

for (int i = 0; i < 5; i++) {
    if (a[i] > largest) {
        secondLargest = largest;
        largest = a[i];
    } else if (a[i] > secondLargest && a[i] != largest) {
        secondLargest = a[i];
    }
}

if (secondLargest == Integer.MIN_VALUE) {
    System.out.println("Second largest element does not exist.");
} else {
    System.out.println("Second largest element = " + secondLargest);
}
}
}

```

Output-

Enter 5 elements:

10 50 30 40 20

Second largest element = 40

24. Remove duplicate elements from a 1-D array (without using collections).

```

import java.util.Scanner;

public class RemoveDuplicates {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");
        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();

```

```

    }

    // Remove duplicates
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                // Shift elements to the left
                for (int k = j; k < n - 1; k++) {
                    arr[k] = arr[k + 1];
                }
                n--;
                j--;
            }
        }
    }

    System.out.println("Array after removing duplicates:");
    for (int i = 0; i < n; i++) {
        System.out.print(arr[i] + " ");
    }

    sc.close();
}
}

```

input-1 2 3 2 4 1 5
output-1 2 3 4 5

25. Find the sum of the main diagonal and secondary diagonal elements of a square matrix.

```

import java.util.Scanner;

public class DiagonalSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the order of matrix: ");
        int n = sc.nextInt();

        int[][] a = new int[n][n];

        System.out.println("Enter the matrix elements:");
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                a[i][j] = sc.nextInt();
            }
        }
    }
}

```

```

int mainSum = 0;
int secondarySum = 0;

for (int i = 0; i < n; i++) {
    mainSum += a[i][i];
    secondarySum += a[i][n - 1 - i];
}

System.out.println("Sum of main diagonal = " + mainSum);
System.out.println("Sum of secondary diagonal = " + secondarySum);
System.out.println("Total sum of both diagonals = " + (mainSum + secondarySum));

sc.close();
}
}

```

Output-

Enter the order of matrix: 3

Enter the matrix elements:

1 2 3

4 5 6

7 8 9

Sum of main diagonal = 15

Sum of secondary diagonal = 15

Total sum of both diagonals = 30

26. Display the upper and lower triangular parts of a square matrix separately.

```

import java.util.Scanner;

public class TriangularMatrix {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the order of matrix: ");
        int n = sc.nextInt();

        int[][] a = new int[n][n];

        System.out.println("Enter the matrix elements:");
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                a[i][j] = sc.nextInt();
            }
        }
    }
}

```

```

        System.out.println("Upper Triangular Matrix:");
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i <= j)
                    System.out.print(a[i][j] + " ");
                else
                    System.out.print("0 ");
            }
            System.out.println();
        }

        System.out.println("Lower Triangular Matrix:");
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i >= j)
                    System.out.print(a[i][j] + " ");
                else
                    System.out.print("0 ");
            }
            System.out.println();
        }

        sc.close();
    }
}

```

Input-

Enter the order of matrix: 3

Enter the matrix elements:

1 2 3

4 5 6

7 8 9

Output-

Upper Triangular Matrix:

1 2 3

0 5 6

0 0 9

Lower Triangular Matrix:

1 0 0

4 5 0

7 8 9

27. Print only the boundary elements of a matrix.

```
import java.util.Scanner;
```

```

public class BoundaryElements {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();

        int[][] a = new int[rows][cols];

        System.out.println("Enter the matrix elements:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.println("Boundary elements:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                if (i == 0 || i == rows - 1 || j == 0 || j == cols - 1)
                    System.out.print(a[i][j] + " ");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }

        sc.close();
    }
}

```

Output-

Boundary elements:

```

1 2 3 4
5   8
9  12
13 14 15 16

```

28. Search for an element in a 2-D array and display its row & column position.

```

import java.util.Scanner;

```

```

public class Search2DArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();

        int[][] a = new int[rows][cols];

        System.out.println("Enter the matrix elements:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.print("Enter element to search: ");
        int search = sc.nextInt();

        boolean found = false;

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                if (a[i][j] == search) {
                    System.out.println("Element found at row " + (i + 1)
                        + " and column " + (j + 1));
                    found = true;
                }
            }
        }

        if (!found) {
            System.out.println("Element not found.");
        }

        sc.close();
    }
}

```

Output-

```

Enter number of rows: 3
Enter number of columns: 3
Enter the matrix elements:
10 20 30
40 50 60

```

70 80 90

Enter element to search: 50

Element found at row 2 and column 2

29. Sort the elements of each row in a jagged array.

```
import java.util.Arrays;
import java.util.Scanner;

public class JaggedArraySort {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        int[][] a = new int[rows][];

        for (int i = 0; i < rows; i++) {
            System.out.print("Enter number of elements in row " + (i + 1) + ": ");
            int cols = sc.nextInt();

            a[i] = new int[cols];

            System.out.println("Enter elements:");
            for (int j = 0; j < cols; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        // Sort each row
        for (int i = 0; i < rows; i++) {
            Arrays.sort(a[i]);
        }

        System.out.println("Jagged array after sorting each row:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < a[i].length; j++) {
                System.out.print(a[i][j] + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

```
}
```

Output-

Enter number of rows: 3

Enter number of elements in row 1: 4

Enter elements:

40 10 30 20

Enter number of elements in row 2: 3

Enter elements:

90 50 70

Enter number of elements in row 3: 5

Enter elements:

8 3 9 1 6

Jagged array after sorting each row:

10 20 30 40

50 70 90

1 3 6 8 9

30. Check if a given matrix is a sparse matrix (more zero elements than non-zero).

```
import java.util.Scanner;
```

```
public class SparseMatrix {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number of rows: ");
```

```
        int rows = sc.nextInt();
```

```
        System.out.print("Enter number of columns: ");
```

```
        int cols = sc.nextInt();
```

```
        int[][] a = new int[rows][cols];
```

```
        int zeroCount = 0;
```

```
        int nonZeroCount = 0;
```

```
        System.out.println("Enter the matrix elements:");
```

```
        for (int i = 0; i < rows; i++) {
```

```
            for (int j = 0; j < cols; j++) {
```

```
                a[i][j] = sc.nextInt();
```

```
                if (a[i][j] == 0)
```

```
                    zeroCount++;
```

```
            else
```

```
                nonZeroCount++;
```

```

    }
}

System.out.println("Number of zero elements = " + zeroCount);
System.out.println("Number of non-zero elements = " + nonZeroCount);

if (zeroCount > nonZeroCount)
    System.out.println("The matrix is a sparse matrix.");
else
    System.out.println("The matrix is not a sparse matrix.");

    sc.close();
}
}

```

Output-

Enter number of rows: 3

Enter number of columns: 3

Enter the matrix elements:

0 0 3

0 5 0

7 0 0

Number of zero elements = 6

Number of non-zero elements = 3

The matrix is a sparse matrix.

